

A Visual Guide to Coordination Compounds

UNDERSTANDING LIGANDS

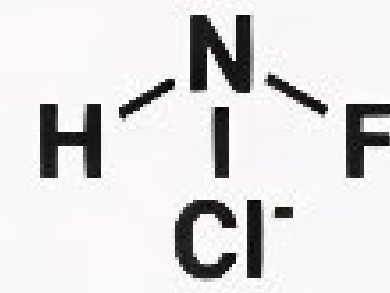
WHAT IS A LIGAND?

Ion/molecule that donates an electron pair to central metal (Coordinate bond)

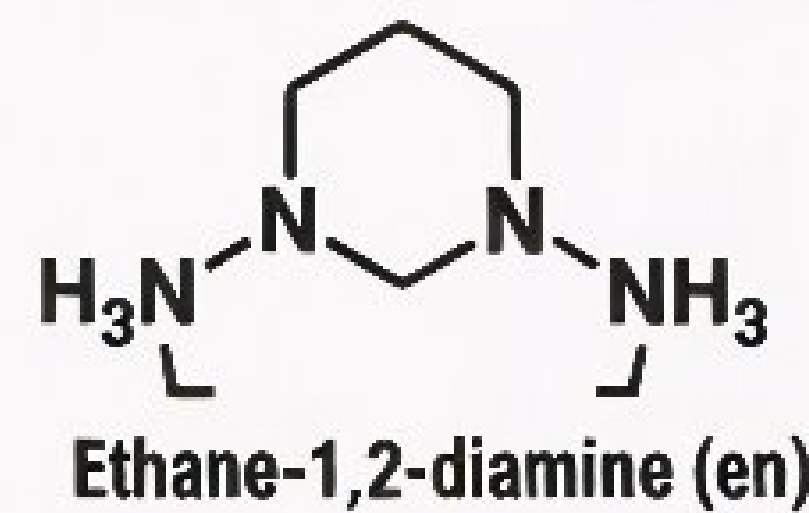
DENTICITY:

Number of Donor Atoms per Ligand

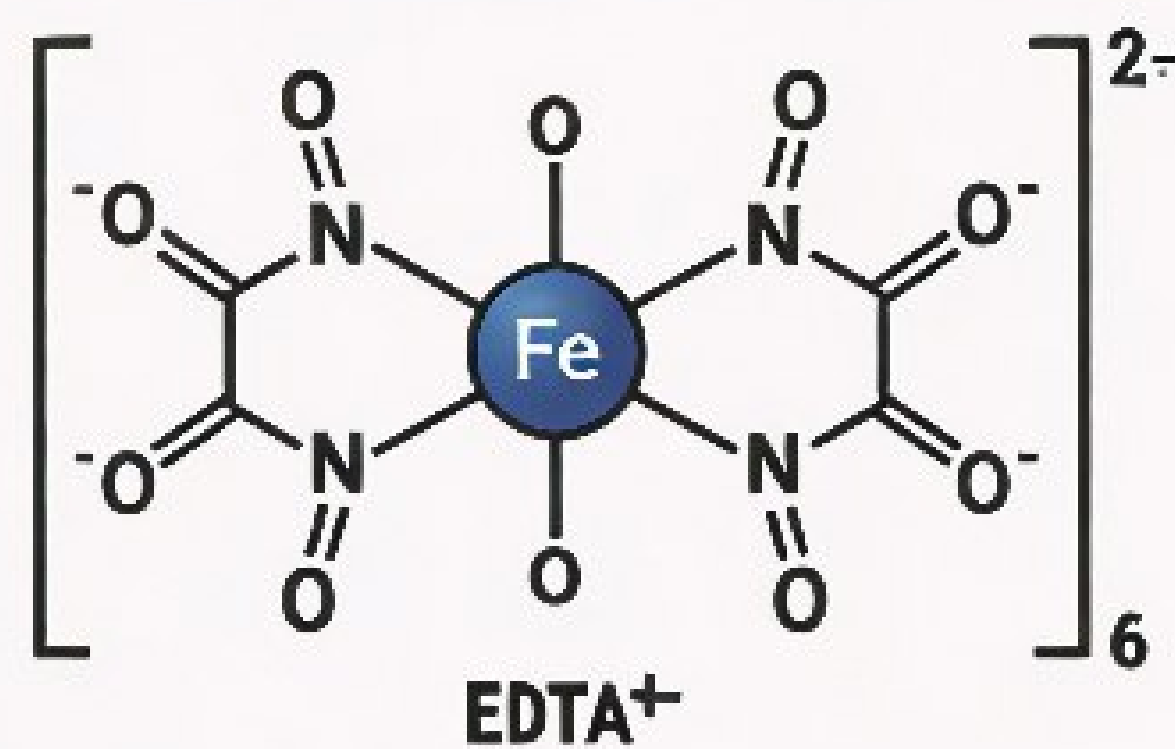
UNIDENTATE (1)



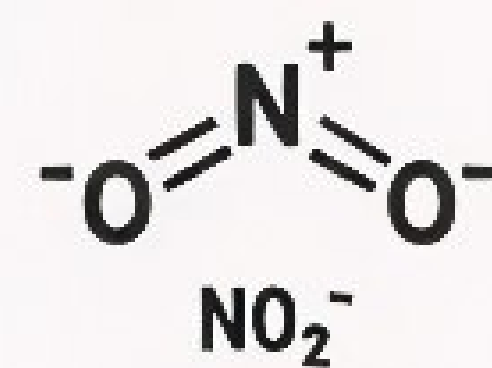
BIDENTATE (2)



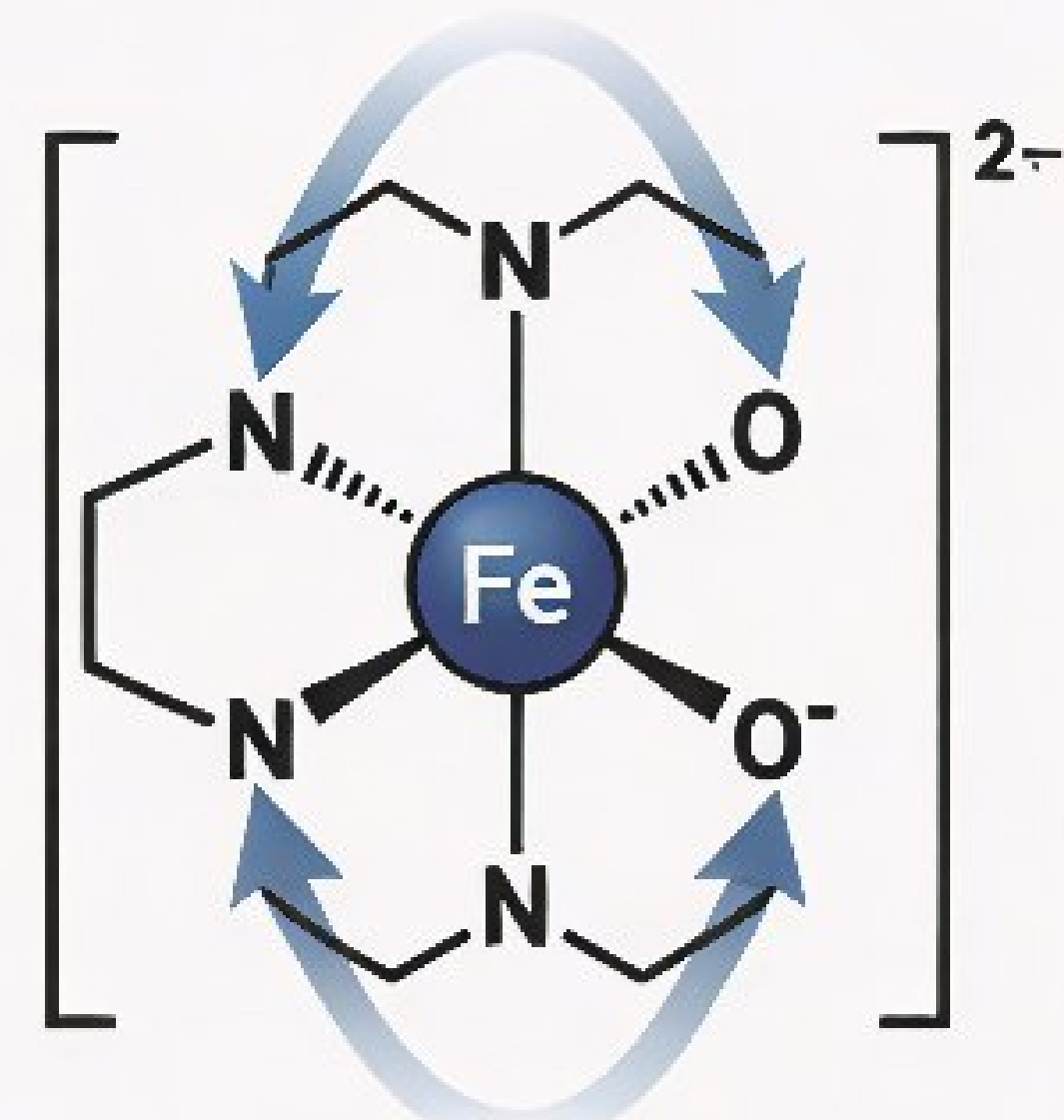
POLYIDENTATE (>2)



AMBIDENTATE



CHELATION: THE "CLAW" EFFECT



Increased Stability

THE ANATOMY OF A COORDINATION COMPOUND



COORDINATION
SPHERE
(Non-ionizable)

COUNTER
ION
(e.g., Cl_4)

LIGANDS
(e.g., CN^-)

LIGANDS
(e.g., CN^-)

CENTRAL METAL ATOM/ION
(e.g., Fe)

COORDINATION
NUMBER:
Total Ligands
Directly Attached
(6)

IUPAC NAMING RULES

HOW TO NAME

1. CATION NAME
2. LIGAND NAME(S)
3. METAL NAME
4. METAL OXIDATION STATE (Roman Numerals)

- STEP 1: FIND OXIDATION STATE**
Set metal as 'X', sum known charges to solve
- STEP 2: NAME LIGANDS**
Alphabetical; use di-, tri-, etc.; use bis-, bis- for complex ligands
- STEP 3: NAME CENTRAL METAL**
Normal name for cation/neutral; '-etc' suffix for anion, e.g., Ferrate
- STEP 4: ASSEMBLE NAME**
e.g., $(\text{Co}(\text{NH}_3)_4\text{Cl}_2)\text{Cl}_2 \rightarrow$ Pentaaquacobalt(III) chloride

COMMON NAMES		ANIONIC SPHERE METAL	
Cf = Chlorido	NH ₃ = Ammine	Fe = Ferrate	Ag = Argentate
CN = Cyanido	H ₂ O = Aqua	Pd = Palladate	An = Aurate
NH = Cannine	CO = Carbonyl	Ag = Argentate	Ni = Nickelate

ISOMERISM IN COORDINATION COMPOUNDS

STRUCTURAL ISOMERS (Different Bonds)

IONIZATION
Debolos consequences of electron sets on inucial bonds



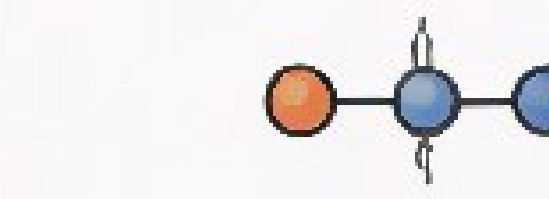
LINKAGE
Hados mananbole ovoiditar coordination or counnritment



HYDRATE/SOLVATE
Dewaes comes or mudrate/solves to central nssnules

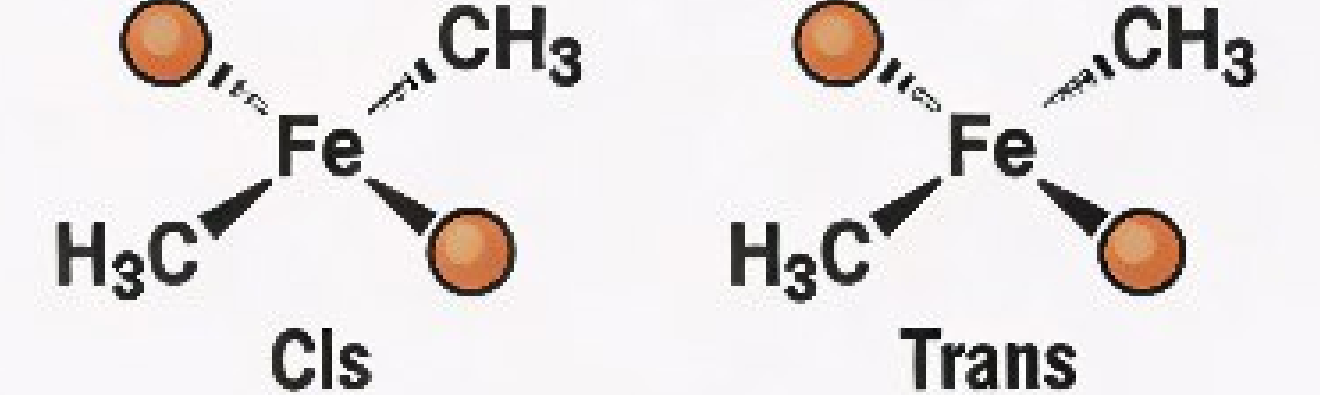


COORDINATION
Complex unnsenmated motor in non supiton spetal arrangement



STEREODISOMERS (Same Bonds, Different Spatial Arrangement)

GEOMETRIC (Cis/Trans)

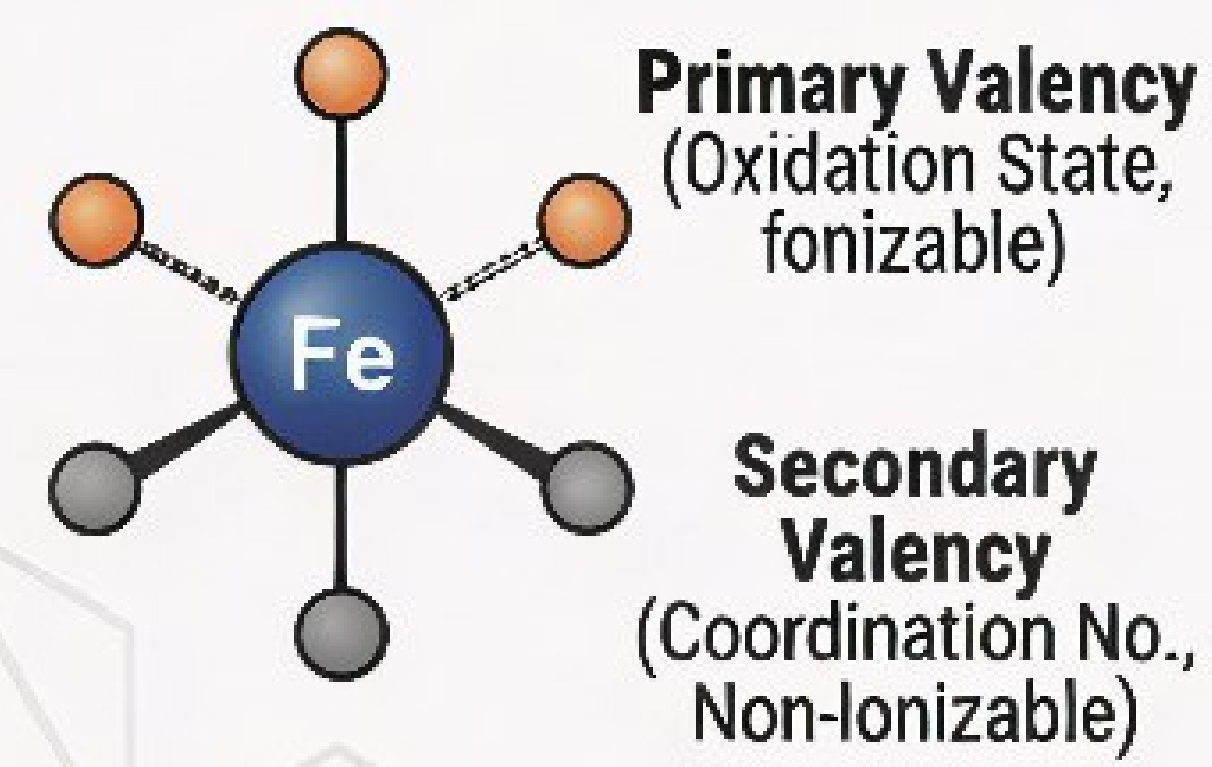


OPTICAL

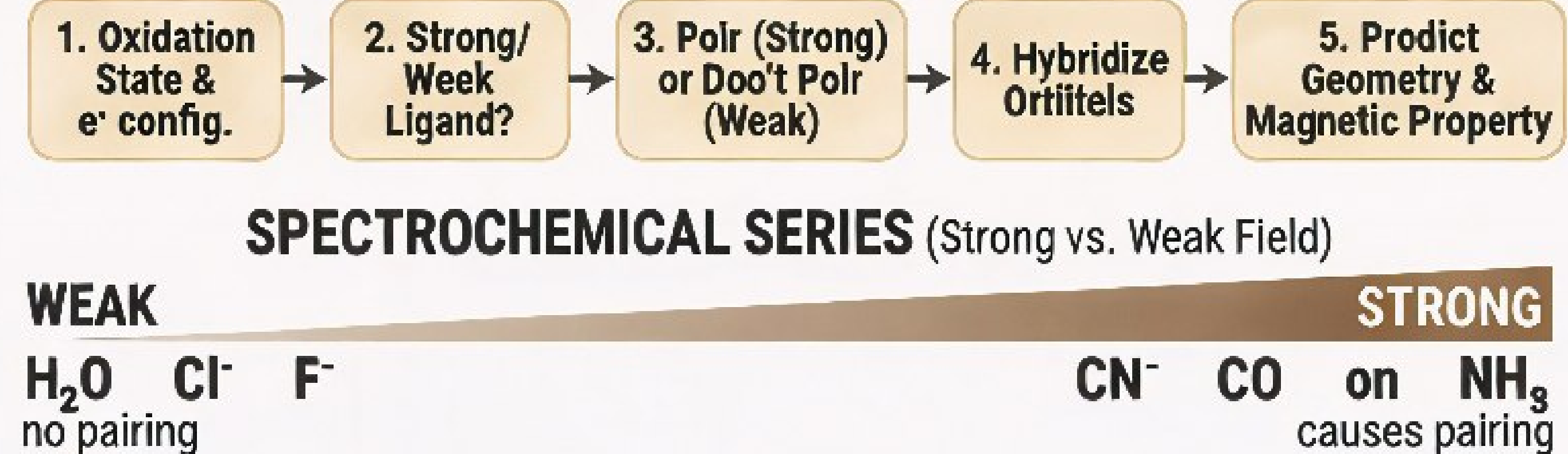


MAJOR BONDING THEORIES

1. WERNER'S THEORY

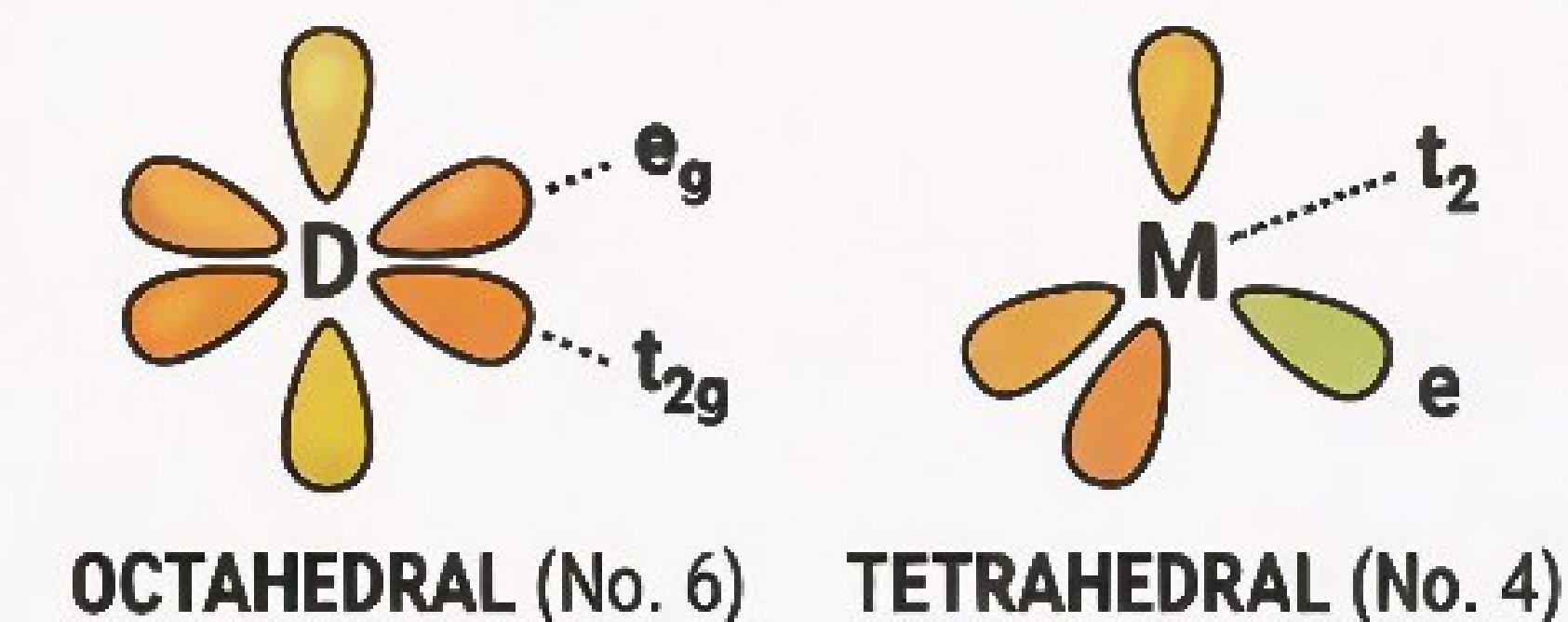


2. VALENCE BOND THEORY (VBT)

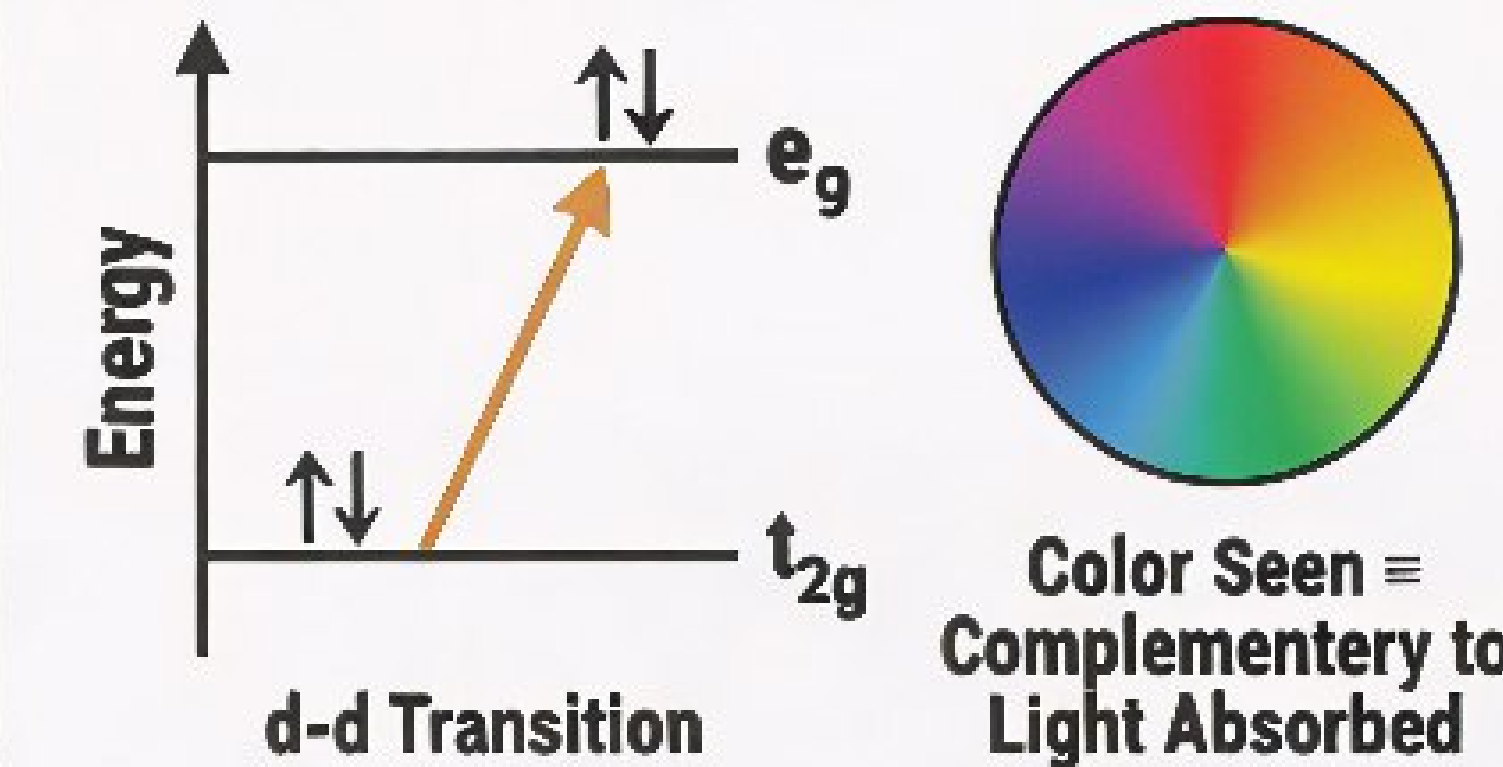


3. CRYSTAL FIELD THEORY (CFT)

D-ORBITAL SPLITTING



THE ORIGIN OF COLOR



SYNERGIC BONDING IN METAL CARBONYLS

